

Fringe Benefits

Production and Amplification of Misinformation on Social Media

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Outline

- 1 Misinformation in a News Cycle
- 2 Model Illustration and Literature
- 3 Reduced-Form Evidence
- 4 A Model of Follower-Stock Competition
- 5 Estimation

Roadmap

1 Misinformation in a News Cycle

- The news cycle
- Measuring misinformation

2 Model Illustration and Literature

3 Reduced-Form Evidence

4 A Model of Follower-Stock Competition

5 Estimation

Monkeypox, 2022: a consequential news cycle, observed end to end

- A **viral disease related to smallpox** — long known, endemic in parts of Africa, usually mild.
- **May 2022**: an unusual **multi-country outbreak** with human-to-human spread outside the endemic region. May 18 — first U.S. case confirmed; WSJ report.
- **Representative** on both counts:
 - ▶ **The cycle** — clean and bounded: the first burst fades inside ten weeks, end to end in our data.
 - ▶ **The misinformation** — COVID-era conspiracy and disinformation templates (Gates, bioweapon, vaccine-harm), combined with heightened uncertainty about a potentially new disease.



DealBook/

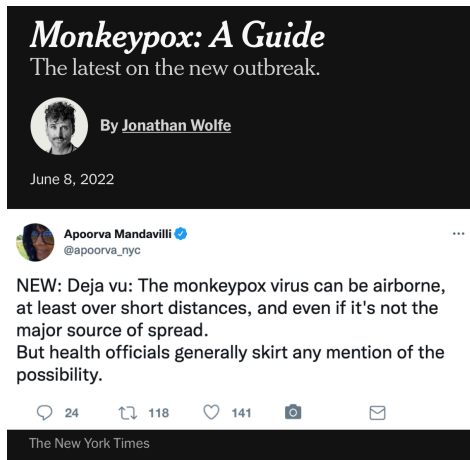
DEALBOOK NEWSLETTER

The strange, swift spread of monkeypox

In his first public remarks on the outbreak, President Biden says that monkeypox is concerning and that “if it were to spread, it would be consequential.”

By [Andrew Ross Sorkin](#), [Vivian Giang](#), [Stephen Gandel](#),
[Lauren Hirsch](#), [Ephrat Livni](#) and [Jenny Gross](#)

Our setting: Viral Tweets on Twitter



Twitter, 2022. A producer posts a tweet; others **like**, **retweet**, **reply** it, and **follow** the authors.

The labeled corpus: 850 tweets · 200 misinfo · 575 accounts

TOPIC monkeypox keywords or hashtags (mpox, vaccine names, etc.)

WINDOW original tweets created between May–Jun 30, 2022

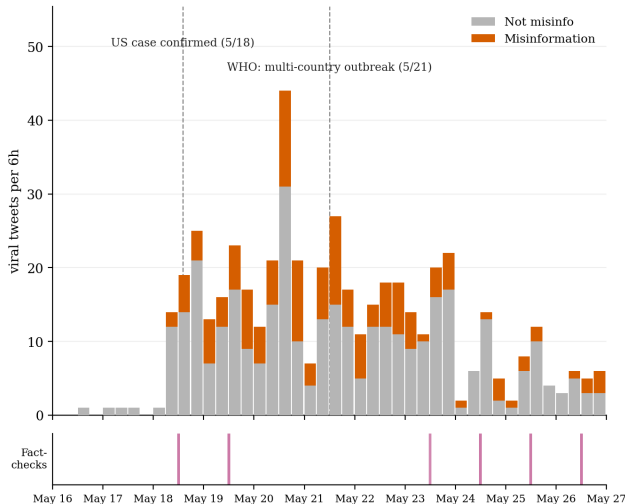
VIRALITY accumulated >1,000 retweeters *or* >100 repliers / quoters (by July 20)

FIELDS author, post time, text & media; cumulative number of retweets, repliers, quoters

LABELS **misinformation** by **archetype** × **topic**, by hand

The onset of the news cycle, May 16–26

Calendar time, 6-hour bins; 503 viral tweets in the window



- **Misinformation arrives on day one** and its share climbs while facts are unsettled: 38% of viral tweets by 5/21 — the **peak coincides with the WHO declaration**.

- **Fact-checks lag** — and the loudest claims escape entirely: 27 dated debunks in the window, most landing *after* the misinfo peak.

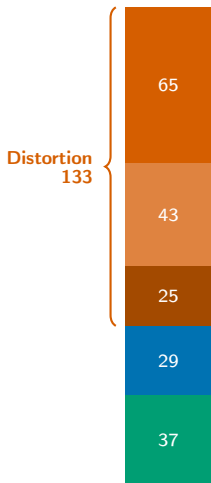
- ▷ **Airborne:** viral 5/18, unresolved 6/6, *false* only 8/5

▷ by content class

▷ weighted by engagement

Misinformation comes in three classes

Five archetypes; Distortion — deliberate — dominates



archetype counts
(a few tweets carry two)

Distortion — false by design:

- **Fringe theories**
- **Sensationalization**
- **Fabricated sources**

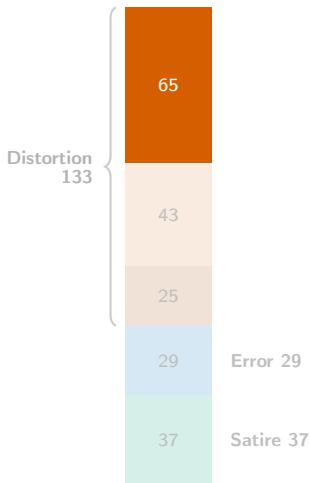
And two smaller classes:

- **Error** — honest slip up, or a wrong forecast
- **Satire** — false on its face, as a joke

► archetypes & examples

Misinformation comes in three classes

Five archetypes; Distortion — deliberate — dominates



archetype counts

(a few tweets carry two)

Fringe theories (65)

the outbreak is a *plot*: bioengineered, or planned by Gates / the WHO / governments.

@TheoFleury14 · May 23, 2022

Monkeypox is the cover up for shingles side effects from the jabs, **pass it on.**

1,037 retweets · 166 replies · 80 quotes

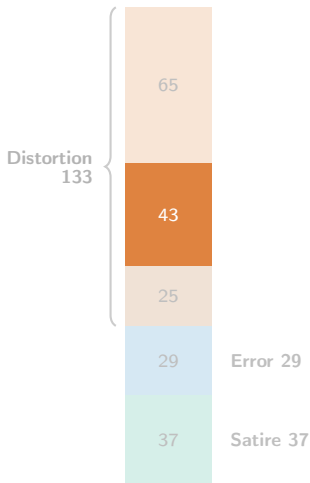
@calvinrobinson · May 20, 2022

Last year: Global Leaders met in Munich to plan responses to a monkeypox outbreak. Bill Gates warned governments to prepare for a smallpox pandemic. ... the same vaccine, **which is ready and good to go.**

1,529 retweets · 319 replies · 374 quotes

Misinformation comes in three classes

Five archetypes; Distortion — deliberate — dominates



archetype counts

(a few tweets carry two)

Sensationalization (43)

a *real* fact stretched to alarm: true cases, catastrophic spin.

@DrEricDing · May 18, 2022

Crap—Monkeypox is likely **aerosol airborne**. Study indicates it is aerosol stable for up to **90 HOURS**...

3,423 retweets · 663 replies · 3,779 quotes

@DailyMailUK · May 30, 2022

Warning to **ABSTAIN from sex** as monkeypox cases jump 70%

59 retweets · 208 replies · 243 quotes

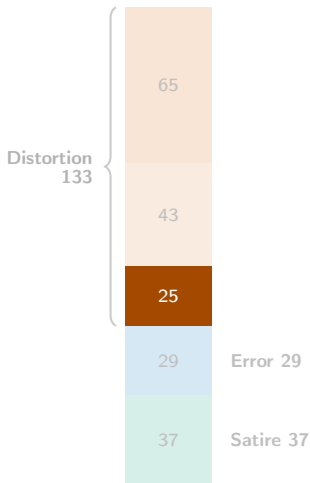
@EnemyInAState · May 19, 2022

*****BREAKING***** ... PARIS DECLARES A SUSPECTED CASE: main route of infection is 'respiratory'

138 retweets · 23 replies · 113 quotes

Misinformation comes in three classes

Five archetypes; Distortion — deliberate — dominates



archetype counts

(a few tweets carry two)

Fabricated sources (25)

borrowed or invented authority: a discredited expert, a relabeled photo.

@backtolife_2022 · May 21, 2022

Dr. Robert Malone about monkeypox [video]

5,435 retweets · 247 replies · 3,188 quotes

@AZmilitary1 · May 27, 2022

Russia Ministry of Defense published photos as evidence of existence of **bio-laboratories in Nigeria**, related to monkey smallpox spreading

283 retweets · 1 replies · 393 quotes

The canonical definition — “intentionally and verifiably false” articles (Allcott & Gentzkow 2017) — reaches **only this archetype**: it excludes conspiracy, slant, error and satire.

Two ways to label misinformation

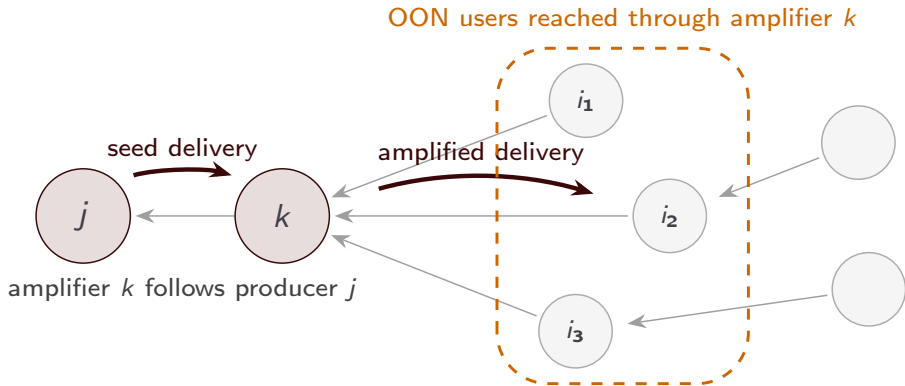
	Match to a fact-check	Ours: adjudicate directly
QUESTION	"has this been debunked by a fact-checking article?"	"is this true — at the time of posting?"
OUTPUT	0/1	a vector : archetype $\times$ topic
SCALE	millions of cascades	one cycle
EVIDENCE	a fact-check URL in the replies (Vosoughi et al. 2018); the union of Snopes + PolitiFact lists (Allcott & Gentzkow 2017)	the record <i>as of the post date</i> ; trail kept
DELETED	unlabelable	Wayback + reverse-image forensics

Matching misses most of the misinformation and is biased: recall tops out at 0.34 even for a human doing exhaustive lookup; 66% of misinfo — especially **sensationalization** (no false sentence to check) and **error** (nobody debunks a wrong forecast) — has no findable fact-check.

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The follow graph alone determines who can reach whom



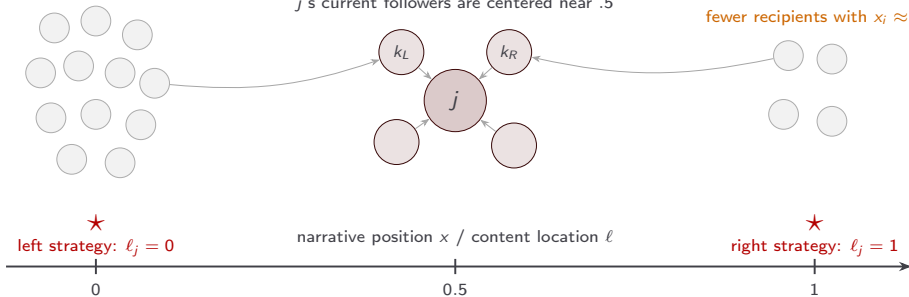
Gray arrows are follow links; thick arrows are content delivery. Content travels against the follow links: producer j reaches follower k , and k 's amplification carries j 's content to some of k 's followers who are OON to j .

Network position and content location are different objects

many recipients with $x_i \approx 0$

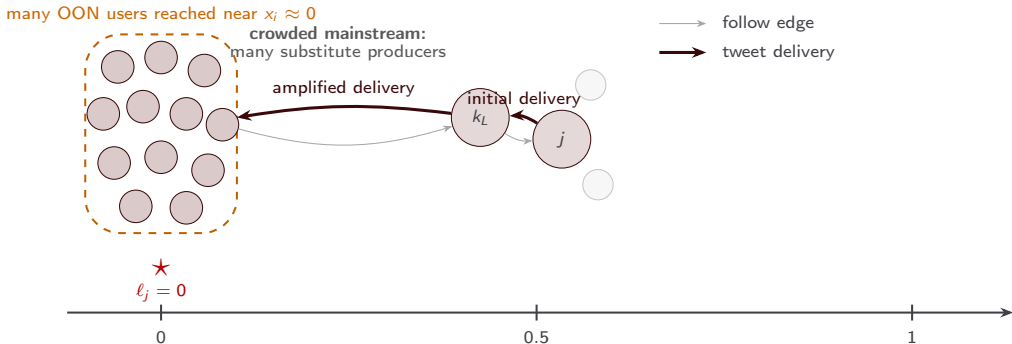
j 's current followers are centered near .5

fewer recipients with $x_i \approx 1$



Network ties determine which viewers an amplifier can expose. The location spectrum determines which viewers value the content. The same producer can use different followers to propagate toward a crowded mainstream location or a thinner fringe location.

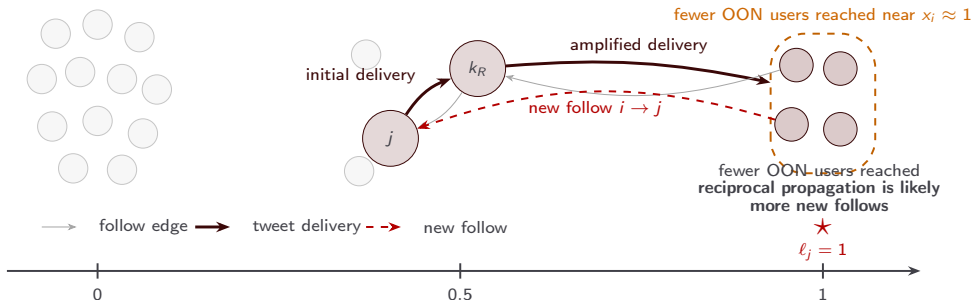
Left strategy: mainstream competition can induce sensationalization



Crowded mainstream: higher potential reach, but higher competition; heavier content investment is needed to generate OON follows or amplification.

Cunningham (2023): quality races down the engagement ranking · Martin, Vasserman & Pfiffer (2026): structural news demand — the engagement-welfare wedge.

Fringe location: fewer views, more follower growth



► Literature

Fewer views, but stronger location preference and reciprocity raise follow conversion; recruits join j 's future distribution network.

Acemoglu, Ozdaglar & Siderius (2024): echo-chamber sharing, veracity exogenous · Gong (2021): niche launch recruits "zealot fans" · Filippas et al. (2025): reciprocity — follows bartered for follow-backs.

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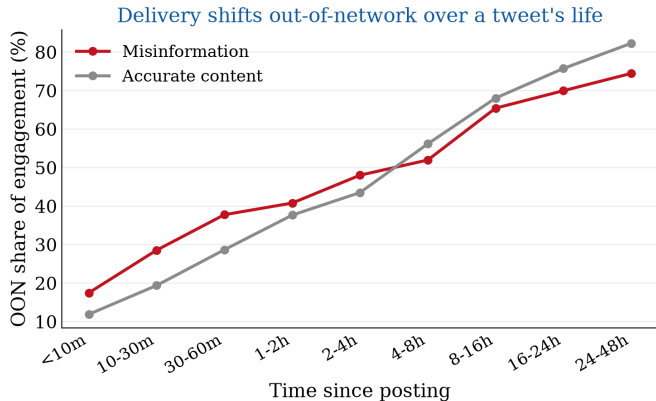
Event-level engagement data and the follow graph

The most-viral tweets, with their full engagement and follow traces

- **Minute-level panel to investigate how the most viral tweets amplified:** we **zoom in on the 71 most-viral tweets** (19 misinfo) of the 850 labeled — all clear the viral bar ($>1,000$ retweeters, or >100 repliers or quoters), and the panel is its top slice (median 1,259 retweeters / 1,654 quoters vs 198/147 across the corpus) — carrying complete event-level data.
- **Engagement panel** (the most comprehensive API firehose, covering 20% of all users): 193,685 timestamped like / retweet / reply / quote actions by 67,670 users.
- **Full follow / unfollow graph** with real dates: each user's follow *state at the moment of engagement* (833,817 edges).

Delivery: in-network first, out-of-network after

Event-level timestamps, first 48 hours after posting

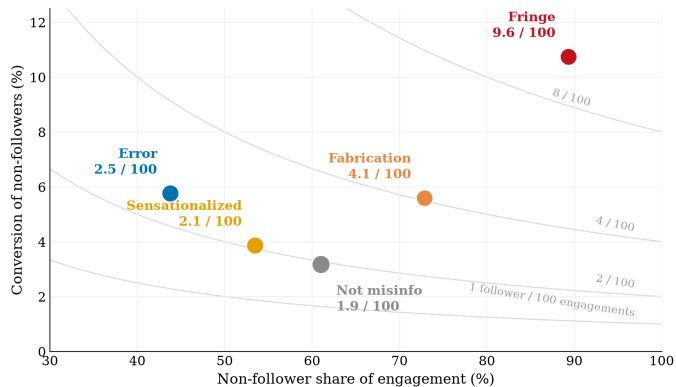


- **First 10 minutes:** ~88% of engagement is by users who **already follow the author** — the seed audience is in-network.
- **By 24–48h the mix inverts:** in-network falls to 18%–26% of engagement.

OOO = out-of-network (engagement by a user not following the author)

Fringe benefits: the follow-conversion premium

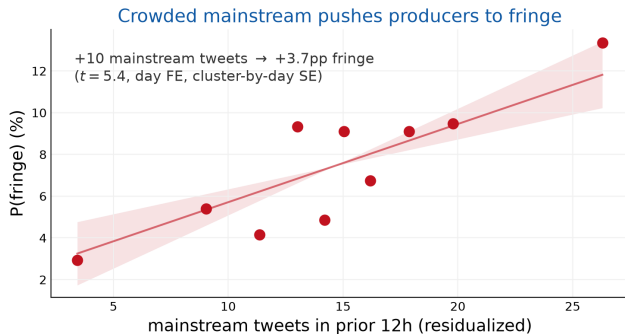
$$\underbrace{\text{new followers per engagement}}_{\text{follower growth}} = \underbrace{P(\text{engager is a non-follower})}_{\text{OON reach / IN amplification}} \times \underbrace{P(\text{follow} \mid \text{non-follower})}_{\text{conversion}}$$



- Fringe is high on *both* margins: 89% of its engagement are non-followers *and* 10.8% of them convert.
- 9.6 new followers per 100 engagements — vs. 1.9 for accurate content ($\sim 5\times$).

Why: congestion and competition at the mainstream

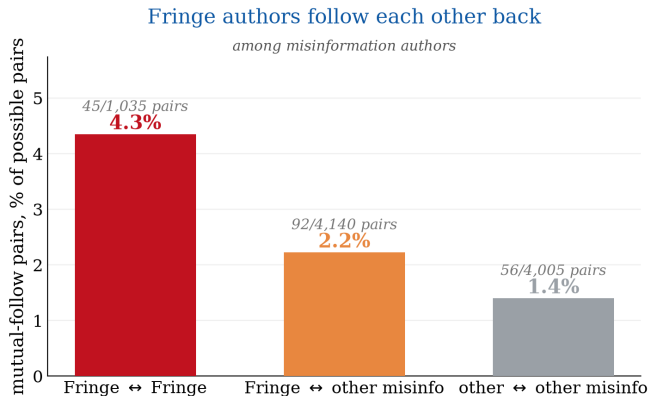
Producers' location choice responds to crowding at the mainstream



- Tweet-level LPM, 646 tweets: **+10 mainstream tweets in the prior 12h $\rightarrow$ +3.7pp fringe** ($t = 5.4$), conditional on the author's follower stock and day fixed effects.
- **Placebo:** trailing *misinfo* congestion does not push to fringe ($t = -0.6$) — it is crowding at *truth*, not activity in general.

Why: fringe producers follow each other back

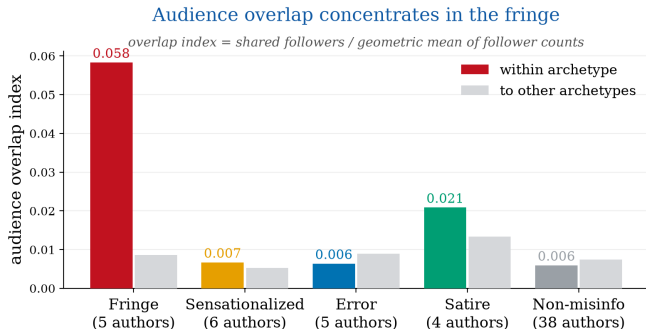
Reciprocal follows among misinfo authors, from their own following lists



- Fringe author pairs are **mutual follows** at 4.3% of possible pairs — 3.1× the rate among other misinfo authors (1.4%).
- 27 protected accounts' follow lists are unobservable and excluded from both numerator and denominator.

Why: fringe audiences overlap

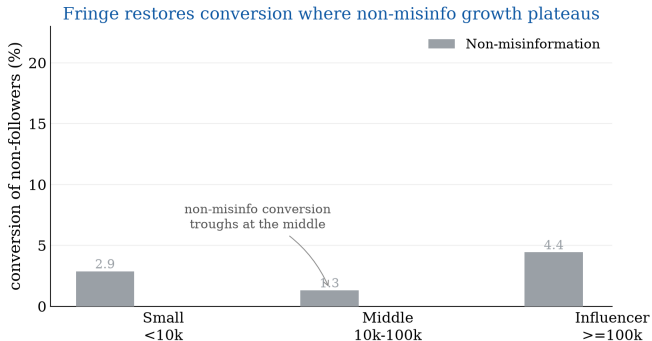
Pairwise audience overlap in the 746,325-follower graph



- Fringe authors share followers with *each other* at 0.058 vs 0.009 with any other archetype (6.8 $\times$).
- No other archetype clusters like this: within-group overlap is flat everywhere else in the network.

Who: the middle squeeze

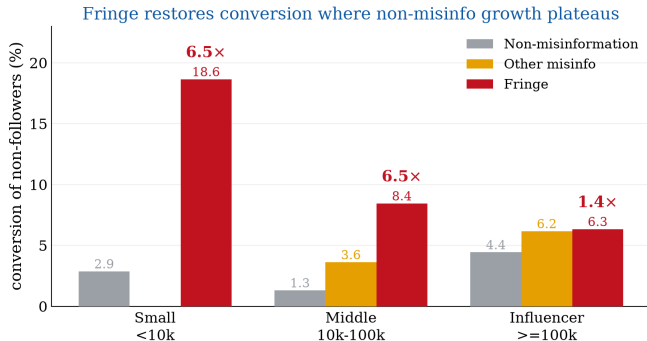
Non-misinformation conversion is U-shaped in audience size — the middle is the trough



- **Middle accounts grow slowest:** they convert non-followers at 1.3% on non-misinformation — below small (2.9%) and influencers (4.4%) — consistent with early niche-driven growth plateauing at this size (Gong 2021).
- Fringe lifts the strugglers, not the strong: conversion rises 6.5× for small and 6.5× for middle accounts, but only 1.4× for influencers who already convert broadly.

Who: the middle squeeze

Non-misinformation conversion is U-shaped in audience size — the middle is the trough

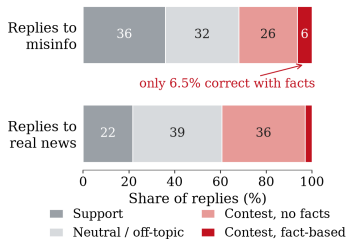


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Self-correction I: debate contests but rarely corrects

Replies of misinformation are more supporting

Replies are 15% of all first-hour engagement events on misinfo tweets (vs 10% on accurate content)



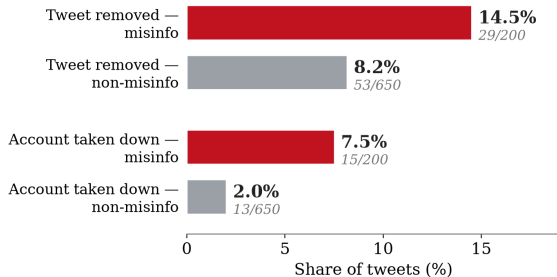
- Misinfo draws *more* early replies: replies are 15% of all engagement events in a misinfo tweet's first hour, vs 10% for accurate content.
- Misinfo replies are **more supportive** (36% vs 22%) and **less contesting** (32% vs 39%) than replies to non-misinformation tweets.
- Only 6.5% are fact-based corrections, and contestation is buried: 2.8% of the likes on misinfo replies.

► like-weighted shares

Self-correction II: deletion and takedown are rare

Deletion and takedown exist, target misinfo, and stay rare

85.5% of misinfo tweets stay live; takedown is nearly a subset of removal (27 of 28 enforced tweets also removed)



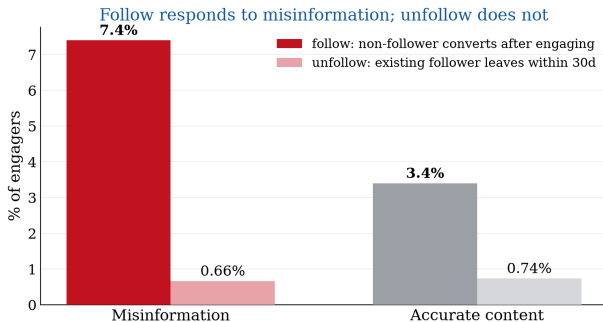
Content gone; self-removal vs platform removal not separable in the data.

- **Tweet removal** is the modal margin yet small: 14.5% of misinfo vs 8.2% of accurate content.
- **Account takedown** is rarer still — 7.5% vs 2.0% — and nearly a subset of removal.
- Net of all removal, 85.5% of misinfo tweets **stay live**.

“Removed” = content gone; self-deletion vs platform removal is not separable — no reason code.

The weak lever: unfollow is low and misinfo-invariant

Existing followers' unfollow decisions, 30 days after each engagement

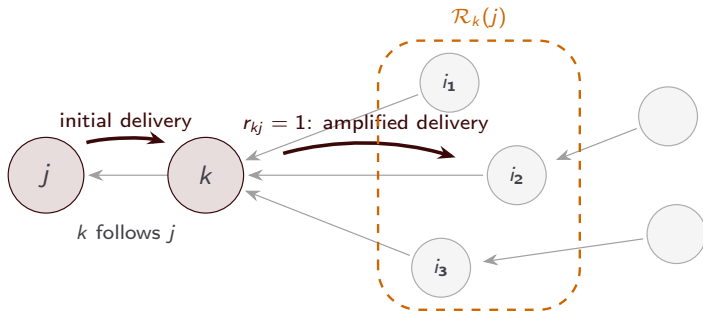


- **Unfollow is rare:** of followers who engage a tweet, 0.66% unfollow the author within 30 days after misinfo vs 0.74% after accurate content — no misinfo penalty (fringe: 0.63%).
- **Asymmetric margins:** the follow response to misinfo (7.4% vs 3.4%) is $\sim 11\times$ the unfollow response — audience gains are not given back.
- **External fact-checks trigger ≈ 0 unfollowing** — the platform delivers no corrections to the audience that followed.

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Refresher: amplification maps the follow graph into OON reach



Gray arrows are follow links; thick arrows are content delivery.

Amplification

$r_{kj} \in \{0, 1\}$ records whether follower k amplifies producer j .

One branch

$\mathcal{R}_k(j) \equiv \{i : i \rightarrow k, i \nleftrightarrow j\}$: users who follow k and have no tie with j .

Realized OON reach

$\mathcal{O}_j(r) \equiv \bigcup_{k:r_{kj}=1} \mathcal{R}_k(j)$: the union of branches activated by amplification.

The producer state is stock, follower locations, and reach

Producer j 's state is

$$z_j = (F_j, \mu_j^F, \mathcal{O}_j(\cdot)) .$$

$\mathcal{O}_j(\cdot)$ maps an amplification profile r into the set of OON users reached.
 μ_j^F is the distribution of follower locations x_i .

1. Producer chooses (ℓ_j, a_j) given rivals' policies.
2. Current followers form the initial seed and may amplify or unfollow.
3. Amplification activates $\mathcal{R}_k(j)$; reached OON users may follow or amplify until $(r_{kj}, \mathcal{O}_j(r))_k$ reaches a fixed point.
4. New follows and unfollows update z_j' .

Notation

F_j : follower stock.

μ_j^F : distribution of follower locations.

$\mathcal{O}_j(\cdot)$: graph reach operator.

State versus outcome

$\mathcal{O}_j(\cdot)$ is the network state; $\mathcal{O}_j(r)$ is its within-tweet output. It retains which amplifier reaches which users, so no separate second-degree-reach statistic or follower-size attribute is needed in μ_j^F .

Two choices $\times$ two channels

Choice	Content channel	Reciprocity channel
k amplifies j	consumption	j follows $k \longrightarrow$ future j amplifies k
i follows j	future i amplifies j (includes future consumption)	j follows $i \longrightarrow$ future j amplifies i

Roles

j : source; k : amplifier; i :
OON recipient

k follows j ; i follows k

i and j do not follow each
other

follows = follow;
amplifies = amplify

Attribution

A plain retweet lets an
OON viewer find the
credited source j (or the
original author), not the
current intermediary k .
Quote/reply visibility is
an extension.

Amplification Choice: r_{kj}

A reached user k chooses whether to amplify producer j . Let c_k^A be k 's amplification cost and $\varepsilon_{kj}^A \sim \text{Logistic}(0, 1)$ the choice shock.

$$U_{kj}^A(z_k) = \underbrace{a_j - \theta_d |x_k - \ell_j|}_{\text{current consumption realized with amplification}} - \underbrace{c_k^A}_{\text{amplification cost}} + \underbrace{\beta \left[1_{k \leftrightarrow j} P_{kj}^A(k \leftarrow j | k \leftrightarrow j) + 1_{k \rightarrow j} P_{kj}^A(k \leftrightarrow j | k \rightarrow j) \right]}_{\text{amplification-triggered follow-back CCP times follower-stock option value}} \left[V_k(z_k^{+j}) - V_k(z_k) \right].$$

$$r_{kj} = 1\{k \text{ is reached}\} 1\{U_{kj}^A + \varepsilon_{kj}^A \geq 0\}, \quad \Pr(r_{kj} = 1 | k \text{ reached}) = \Lambda(U_{kj}^A).$$

Reminders

- All payoffs are in follower equivalents ($m \equiv 1$).

$P_{kj}^A(s' | s)$: source j 's follow-back CCP after k amplifies

$k \leftrightarrow j \rightarrow k \leftarrow j$: no tie to one-way

$k \rightarrow j \rightarrow k \leftrightarrow j$: one-way to mutual

1_s : current-state indicator

z_k^{+j} : k 's state after j follows k

β : discount factor

ε_{kj}^A : logistic choice shock

r_{kj} : observed amplification action

► Amplification modes

► State evolution

Follow/unfollow Choice: $i \rightarrow j$ / $i \nrightarrow j$

An OON user i may follow producer j .

$$W_{ij} = \underbrace{\alpha_i}_{\text{recipient baseline}} + \underbrace{\beta \mathbb{E} \left[\left(U_{ij}^A + \varepsilon_{ij}^A \right)_+ \mid i \rightarrow j \right]}_{\text{future amplification opportunities, including consumption}} + \underbrace{1_{i \leftrightarrow j} P_{ij}^F (i \leftrightarrow j \mid i \leftrightarrow j) \beta \left[V_i(z_i^{+j}) - V_i(z_i) \right]}_{\text{follow-triggered reciprocity: } j \text{ follows } i \text{ and may later amplify } i}.$$

The corresponding transition probabilities are

$$q_{ij} \equiv \Pr \left(W_{ij} + \varepsilon_{ij}^F \geq 0 \mid i \in \mathcal{O}_j(r), i \leftrightarrow j \right),$$

$$u_{ij} \equiv \Pr \left(W_{ij} + \varepsilon_{ij}^U < -c_U \mid i \rightarrow j \right).$$

Notation

W_{ij} : value of following j

P_{ij}^F : no tie $\rightarrow$ mutual after follow-back

q_{ij} : OON follow probability

u_{ij} : current-follower unfollow probability

$\varepsilon_{ij}^F, \varepsilon_{ij}^U$: choice shocks

$(x)_+ = \max\{x, 0\}$

c_U : unfollow friction

Eligible users

q_{ij} : reached no-tie pairs;
 u_{ij} : current followers, including mutual pairs (Overgoor, Benson & Ugander, 2019). Shocks remain open here; Section 5 uses logit for inversion.

► Amplification modes

► State evolution

Producer problem: jointly choose location and appeal

Let z denote the profile of all account states. Taking rivals' Markov policies as given, producer j solves

$$V_j(z) = \max_{\substack{\ell \in \{0,1\} \\ a \geq 0}} \left\{ \underbrace{F_j}_{\text{current follower-stock profit}} - \underbrace{c_j(\ell) - c_a(a)}_{\text{location and appeal costs}} + \underbrace{\beta \mathbb{E}[V_j(z') \mid z, \ell, a]}_{\text{expected continuation value}} \right\}.$$

Conditional on the optimal location, an interior appeal choice satisfies

$$c'_a(a_j^*) = \beta \frac{d}{da} \mathbb{E}[V_j(z') \mid z, \ell_j^*, a] \Big|_{a=a_j^*}.$$

Because location is discrete, j compares the two locations directly:

$$\ell_j^* \in \arg \max_{\ell \in \{0,1\}} \left\{ -c_j(\ell) - c_a(a_j^*(\ell)) + \beta \mathbb{E}[V_j(z') \mid z, \ell, a_j^*(\ell)] \right\}.$$

Notation

z : profile of account states

β : discount factor

The expectation integrates the follow-choice shocks and pair-level state transitions.

Optimality

The total derivative includes how appeal changes amplification, following, unfollowing, reciprocity, and graph reach. Location uses a direct comparison, not an FOC.

Equilibrium links pair-level transitions to the next state

A Markov equilibrium satisfies five conditions:

1. **Follow optimality:** q_{ij} and u_{ij} are the probabilities implied by W_{ij} and the choice shocks.
2. **Amplification optimality:** reached k chooses $r_{kj} = 1\{U_{kj}^A + \varepsilon_{kj}^A \geq 0\}$, and $\mathcal{O}_j(r) = \bigcup_{k:r_{kj}=1} \mathcal{R}_k(j)$.
3. **Follow-back CCP consistency:**

$$P_{ij}^F(s' | s) = \Pr(s_{ij,t+1} = s' | s_{ij,t} = s, i \text{ follows } j),$$

$$P_{kj}^A(s' | s) = \Pr(s_{kj,t+1} = s' | s_{kj,t} = s, r_{kj,t} = 1).$$

4. **State-law consistency:** the appendix state-evolution blocks aggregate realized pair-level outcomes into $z'_j = (F'_j, \mu_j^{F'}, \mathcal{O}'_j(\cdot))$. The conditional law in the producer Bellman problem equals the law induced by q, u, r, P^F , and P^A .
5. **Producer optimality:** (ℓ_j, a_j) jointly solve the producer problem, and V_j satisfies its Bellman equation.

Follow state

s is the current follow relation; s' is next period's relation.

CCPs and counterfactuals

Estimate source follow-back policies and plug them into continuation values.

Holding them fixed gives a conditional counterfactual; full equilibrium requires a source payoff and best-response iteration.

Stationarity

A stationary equilibrium also requires an invariant distribution of z .

Model extension: separate entertainment from truthfulness

The baseline scalar a_j is a reduced-form consumption index. Decompose content quality as

$$a_j = (e_j, t_j), \quad e_j = e_j^S + e_j^H, \quad t_j \in [0, 1],$$

where e_j is entertainment and t_j is truthfulness. On the amplification payoff, replace the consumption term with

$$\underbrace{e_j + \theta_t t_j}_{\text{consumption value}} - \underbrace{\theta_d |x_k - \ell_j|}_{\text{location preference}}, \quad \theta_t \geq 0.$$

Producer j chooses the least-cost entertainment mix:

$$c_a(a_j) \equiv \min_{\substack{e_j^S, e_j^H \geq 0 \\ e_j^S + e_j^H = e_j}} \left\{ c_S(e_j^S) + c_H(e_j^H) + c_T(t_j) + \gamma e_j^S t_j \right\}, \quad \gamma > 0.$$

Truthful sensationalization is costly; humor or satire can raise entertainment without lowering truthfulness.

Components

e_j^S : sensationalization

e_j^H : humor or satire

t_j : truthfulness

c_S, c_H, c_T : production and verification costs

θ_t : consumption value of truthfulness

γ : cost of combining sensationalization with truthfulness

Interpretation

High entertainment is not misinformation; low t_j is. Competition selects sensationalization or humor. Cunningham (2023).

Extension scope. Demand, cost, and measurement only; the benchmark retains scalar a_j .

Model extension: an amplification reach tax

In the baseline, every branch in $\mathcal{O}_j(r)$ is delivered. Introduce an amplification reach tax $\lambda \in [0, 1]$. For each reachable user, draw a fixed $\xi_{ij} \sim U[0, 1]$ and define

$$\mathcal{O}_j^\lambda(r) = \{i \in \mathcal{O}_j(r) : \xi_{ij} > \lambda\}.$$

Thus $\lambda = 0$ is the baseline and $\lambda = 1$ blocks all OON pass-through. Conditional on r , define total reached users as

$$Y_j^\lambda \equiv F_j + \int 1\{i \in \mathcal{O}_j^\lambda(r)\} dG_i,$$

$$\mathbb{E}[Y_j^\lambda \mid r] = F_j + (1 - \lambda) \int 1\{i \in \mathcal{O}_j(r)\} dG_i.$$

Using the baseline follow and unfollow probabilities, the policy-adjusted expected follower flow is

$$\phi_{ij}^\lambda(r) = \underbrace{q_{ij} 1\{i \in \mathcal{O}_j^\lambda(r)\}}_{\text{taxed OON recruitment}} - \underbrace{u_{ij} 1\{i \rightarrow j\}}_{\text{ordinary unfollowing}}.$$

Policy

λ : fraction of mechanically reachable OON users withheld by the platform.

Timing

The platform applies the tax after amplification. The comparison therefore holds r fixed and varies only OON pass-through.

Equilibrium feedback

An anticipated tax can also change r , q , u , content investment, and ℓ_j through continuation value.

Content censorship and account suspension target different margins

Content censorship can remove a tweet before propagation or suspend its account before the next period. Allow both hazards to depend on content location:

$$h_{\ell}^C = \Pr(\text{tweet deleted before propagation} \mid \ell),$$

$$h_{\ell}^E = \Pr(\text{account suspended before next period} \mid \ell).$$

Let $\mathcal{O}_j^C(r)$ be the reached set after the deletion decision. Tweet deletion reduces current recruitment:

$$\mathbb{E}\left[1\{i \in \mathcal{O}_j^C(r)\} \mid \ell_j\right] = (1 - h_{\ell_j}^C)1\{i \in \mathcal{O}_j(r)\}.$$

Account suspension instead reduces continuation value:

$$V_j(z) = \max_{\ell, a} \left\{ F_j - c_j(\ell) - c_a(a) + \underbrace{\beta(1 - h_{\ell}^E)\mathbb{E}[V_j(z') \mid z, \ell, a]}_{\text{continuation only if the account survives}} \right\}.$$

Tweet lever

h_{ℓ}^C removes current content before it generates OON reach and new follows; existing follower stock remains.

Account lever

h_{ℓ}^E destroys the producer's continuation value and all future use of follower stock.

Location targeting

The baseline sets both hazards to zero. Censorship may instead target fringe content with h_1^C, h_1^E above their mainstream counterparts.

Compare levers at the same misinformation reduction

Let π index a policy built from the levers on the reach-tax and censorship extensions, with $\pi = 0$ denoting the baseline. Let r_{kj}^π be equilibrium amplification and $R_j^\pi \equiv \int r_{kj}^\pi dG_k$ its aggregate. Define a researcher-chosen misinformation weight

$$\chi_j \equiv \omega_\ell \ell_j + \omega_t(1 - t_j), \quad M^\pi \equiv \mathbb{E} \left[\sum_j \chi_j R_j^\pi \right].$$

For each lever, choose its intensity to deliver the same reduction ΔM :

$$M^0 - M^\pi = \Delta M.$$

Let F_j^π be producer j 's follower stock at a common evaluation horizon. Compare

$$L_R^\pi(\Delta M) \equiv \mathbb{E} \left[\sum_j (R_j^0 - R_j^\pi) \right],$$
$$L_F^\pi(\Delta M) \equiv \mathbb{E} \left[\sum_j (F_j^0 - F_j^\pi) \right].$$

Measurement

χ_j : researcher-chosen weight on fringe location or untruthfulness

ω_ℓ, ω_t : nonnegative researcher weights

M^π : weighted misinformation amplification

ΔM : common reduction target

Efficacy

L_R^π : lost amplification

L_F^π : lost follower stock

Report both at the same ΔM ; no platform welfare weights are required.

Model extension: screen by amplification mode

Let $\tau \in \mathcal{M} = \{\text{fav}, \text{RT}, \text{reply}, \text{quote}\}$ index amplification mode, r_{kj}^τ user k 's mode- τ action, and

$$R_j^\tau = \int r_{kj}^\tau dG_k, \quad R_j = (R_j^\tau)_{\tau \in \mathcal{M}}.$$

Let r collect all mode-specific actions. Every observed mode jointly reveals consumption and amplification, but only propagating modes create an OON reach branch:

$$\mathcal{O}_j(r) = \bigcup_{k, \tau: r_{kj}^\tau = 1} \mathcal{R}_k^\tau(j), \quad \mathcal{R}_k^{\text{fav}}(j) = \emptyset.$$

When location or truthfulness is not directly observed, the platform forms

$$\hat{p}_j = \Pr(\ell_j = 1 \text{ or } t_j < \bar{t} \mid R_j), \quad (\lambda_j, h_j^C, h_j^E) = \Gamma(\hat{p}_j).$$

Favorites can therefore inform screening without creating an OON branch. Retweets, replies, and quotes can differ in reach and attribution. Reply-based screening also requires stance.

Modes

Favorite: consumption + amplification, no OON branch

Retweet: consumption + source-attributed propagation

Reply/quote: consumption + mode-specific propagation

Screen

$\hat{p}_j$: posterior probability of fringe location or low truthfulness

$\bar{t}$: low-truthfulness cutoff

Γ : policy rule mapping the posterior into the three levers

Policy comparisons do not require a platform objective

Under the preceding timing and regularity conditions:

1. **Reach monotonicity.** Holding r fixed, for $\lambda' \geq \lambda$,

$$\mathcal{O}_j^{\lambda'}(r) \subseteq \mathcal{O}_j^{\lambda}(r), \quad \mathbb{E}[Y_j^{\lambda'} | r] \leq \mathbb{E}[Y_j^{\lambda} | r].$$

2. **Flow versus stock.** Higher h_{ℓ}^C weakly reduces current OON recruitment without directly destroying F_j ; higher h_{ℓ}^E weakly reduces continuation value and all future use of F_j .
3. **Producer response.** If fringe location and appeal complement effective reach and continuation value, a stronger λ , h_1^C , or h_1^E weakly lowers fringe adoption and optimal appeal.
4. **Value of multimodal screening.** If R_j Blackwell-dominates retweets alone for predicting fringe location or low truthfulness, then at any ΔM the attainable set of (L_R, L_F) weakly expands.

Assumptions

Nested reach thinning for result 1; nonnegative continuation value for result 2; convex appeal cost, single crossing, and a unique optimum for result 3; multimodal policies nest retweet-only policies for result 4.

Interpretation

Amplification taxes and tweet deletion target current flow. Suspension targets durable follower stock. Richer signals can improve the efficacy frontier without a scalar welfare function.

Roadmap

- 1 Misinformation in a News Cycle
- 2 Model Illustration and Literature
- 3 Reduced-Form Evidence
- 4 A Model of Follower-Stock Competition
- 5 Estimation

Estimate choice probabilities first; then recover payoffs

1. **Estimate observed choice probabilities.** Estimate recipient policies q_{ij}, u_{ij} and source follow-back policies $P_{ij}^F(s' | s), P_{kj}^A(s' | s)$ as choice probabilities among eligible users in each current follow state.
2. **Recover the subscription value W_{ij} .** With standard logit shocks,

$$W_{ij} = \log \frac{q_{ij}}{1 - q_{ij}} = \log \frac{1 - u_{ij}}{u_{ij}} - c_U.$$

A useful consistency check is that follow and unfollow imply the same W_{ij} .

3. **Map choices into future states.** Plug P^F, P^A and the mechanical state transition into V , then estimate producer choices. Source follow-back policies enter continuation values, but are not yet inverted into a source payoff.

Recipient choices

q, u are two choices linked by one subscription value W . They should imply the same value.

Source follow-back

P^F, P^A are estimated from reverse-edge transitions and enter future value.

Observed follow and unfollow transitions use one seven-day window

Let t_{ij}^0 be the first clean action episode for account pair (i, j) . Keep one episode per account pair and set $h = 7$ days on both margins.

$$\hat{q}^{\text{obs}}(7\text{d}) = \Pr(i \rightarrow j \text{ within } 7\text{d} \mid i \nrightarrow j \text{ at } t_{ij}^0, \text{ full } 7\text{d follow-up}),$$

$$\hat{u}^{\text{obs}}(7\text{d}) = \Pr(i \nrightarrow j \text{ within } 7\text{d} \mid i \rightarrow j \text{ at } t_{ij}^0, \text{ full } 7\text{d follow-up}).$$

$$\underbrace{\frac{853}{1,701}}_{\text{follow among selected non-followers}} = 50.1\%$$

95% CI=[47.8%, 52.5%]

$$\underbrace{\frac{160}{5,711}}_{\text{unfollow among selected followers}} = 2.8\%$$

95% CI=[2.4%, 3.3%]

Observed transitions for selected actors—not population follow or unfollow probabilities.

Common horizon

One clean episode per account pair. Keep it only if 7 days remain in the relevant event log.

Observed denominators

Follow pairs:

1,705 total; 1,703 at 24h;
1,701 at 7d.

Unfollow pairs:

5,721 total; 5,711 at 7d.

Selection

obs means an observed actor and a source-edge history somewhere in the log. These are calibration moments, not population choice probabilities.

Structural gap

Structural q_{ij} also requires verified OON reach and reached non-followers.

Policy functions map the current state into amplification and follow

Amplification. Among users who receive producer j 's content, let $\mathcal{I}_{kj} \equiv \{k \text{ reached}, z, \ell_j, a_j\}$ denote the observed conditioning information. Then

$$\Pr(r_{kj} = 1 \mid \mathcal{I}_{kj}) = \Pr(U_{kj}^A + \varepsilon_{kj}^A \geq 0 \mid \mathcal{I}_{kj}) = \Lambda(U_{kj}^A),$$

where $\varepsilon_{kj}^A \sim \text{Logistic}(0, 1)$. The systematic payoff combines current consumption, k 's amplification cost, and the source follow-back option.

Follow. Among reached OON users with no tie to j ,

$$q_{ij} = \Pr\left(W_{ij} + \varepsilon_{ij}^F \geq 0 \mid i \in \mathcal{O}_j(r), i \nleftrightarrow j\right).$$

Location preference, future amplification, and follow-back determine W_{ij} ; unfollow provides the second choice probability for the same value.

First estimate these choice probabilities flexibly conditional on state. Then impose the shared payoff structure when recovering model parameters.

Eligible to amplify

All accounts reached before the amplification decision. First-wave current followers are observable; later-wave OON non-actors require impressions or latent-reach integration.

Eligible to follow

Reached OON users satisfying $i \nleftrightarrow j$, including those who do not follow.

State variation

Current follow relationship; follower stock; follower composition; locations; content appeal; follow-back policy.

Timing narrows propagation paths, but cannot recover exposure

Exact reach requires an edge that is missing for arbitrary amplifier k :

$$i \in \mathcal{O}_j(r) \iff \exists k : \underbrace{r_{kj} = 1}_{\text{observed}}, \quad \underbrace{i \rightarrow k}_{\text{missing}}, \quad \underbrace{i \nleftrightarrow j}_{\text{partly observed}}$$

Timing identifies candidate propagators for the same tweet:

$$\mathcal{K}_{ij}(\Delta) \equiv \left\{ k : \underbrace{k \text{ retweets, replies, or quotes at } t_{kj}}_{\text{public propagation}}, \quad 0 < t_{ij}^0 - t_{kj} \leq \Delta \right\}.$$

Timing can summarize when later actions arrive after propagation. It does not show that a particular user was exposed through a particular amplifier.

Symbols

t_{ij}^0 : i 's first clean OON action.

t_{kj}, τ_{kj} : k 's prior propagation time and mode.

Timing support

1,704/1,705 selected non-current account pairs have a prior propagator; median latest lag: 1.7 min.

Within 5 min:
72.0% have candidates;
median 3.0 if positive.

Within 30 min:
91.9% have candidates;
median 12.0.

What timing cannot do

Exact reach needs sampled amplifier follower lists or impressions. Timing never reveals exposed users who take no action.

Three missing denominators limit what we can estimate

1. **Exposure.** We cannot recover exact $\mathcal{O}_j(r)$ without recipient–amplifier edges or impressions. Timing narrows propagation parents but does not recover exposed users who take no action.
2. **Choice probabilities.** Amplification and follow probabilities need users who could choose but did not. Reverse-edge timing is also incomplete, so unsupported follow-back policies are calibrated and reported with sensitivity.
3. **Counterfactual scope.** Holding source CCPs fixed compares policies while assuming follow-back behavior does not change. A full equilibrium response requires a source follow-back payoff.

Identified now

Action times and modes;
current follow state;
selected 7-day transitions;
supported reverse-edge
changes.

Round one

Estimate selected
transitions and supported
CCPs; bound latent reach;
label conditional
counterfactuals.

Round two

Add exposure data or
network reconstruction;
identify supply shifters;
model source follow-back
payoffs for full equilibrium.

Thank you

Appendix: how amplification evolves the state

Condition on the current global state z and amplification profile r . Source j 's OON reached set is

$$\mathcal{O}_j(r) = \bigcup_{k:r_{kj}=1} \mathcal{R}_k(j).$$

Using the transition probabilities from the follow-choice block, define user i 's expected net contribution to j 's follower stock:

$$\phi_{ij}(r) = \underbrace{q_{ij}1\{i \in \mathcal{O}_j(r)\}}_{\text{reached OON user follows } j} - \underbrace{u_{ij}1\{i \rightarrow j\}}_{\text{current follower unfollows } j}.$$

Aggregating across users gives the mean transition:

$$\begin{aligned}\mathbb{E}[F'_j \mid z, r] &= F_j + \int \phi_{ij}(r) dG_i, \\ \mathbb{E}[F'_j \mu_j^{F'} \mid z, r] &= F_j \mu_j^F + \int \phi_{ij}(r) \delta_{x_i} dG_i.\end{aligned}$$

Realized follows and unfollows update the graph and therefore determine $\mathcal{O}'_j(\cdot)$. Amplification may also change k 's state through source follow-back:

$$\begin{aligned}\Pr(z'_k = z_k^{+j} \mid r_{kj} = 1, k \nleftrightarrow j) &= P_{kj}^A(k \leftarrow j \mid k \nleftrightarrow j), \\ \Pr(z'_k = z_k^{+j} \mid r_{kj} = 1, k \rightarrow j, j \nrightarrow k) &= P_{kj}^A(k \leftrightarrow j \mid k \rightarrow j).\end{aligned}$$

Source state

dG_i : population measure

δ_{x_i} : point mass at follower location

Thus $\phi_{ij}(r)$ updates follower stock and its unnormalized distribution.

Amplifier state

Amplification changes k 's state only if source j follows k ; a plain retweet gives k no direct OON followers.

Transition

Estimate each CCP among account pairs in the corresponding current state; externally calibrate unsupported reverse-edge transitions.

◀ Back to choice

Appendix: how following evolves the state

If OON user i follows producer j , j 's follower stock rises by one:

$$F_j^{+i} = F_j + 1,$$
$$\mu_j^{F,+i} = \frac{F_j \mu_j^F + \delta_{x_i}}{F_j + 1}.$$

Let $\mathcal{O}_j^{+i}(\cdot)$ be the reach operator on the graph after adding $i \rightarrow j$. The follow updates j 's state to

$$z_j^{+i} = \left(F_j^{+i}, \mu_j^{F,+i}, \mathcal{O}_j^{+i}(\cdot) \right).$$

Starting from no tie, j may follow i back, creating a second state change:

$$\Pr(z_i' = z_i^{+j} \mid i \text{ follows } j, i \nleftrightarrow j) = P_{ij}^F(i \leftrightarrow j \mid i \nleftrightarrow j).$$

A realized unfollow removes $i \rightarrow j$ and i 's contribution to j 's stock and composition. If the pair was mutual, $j \rightarrow i$ remains.

Direct effect

$i \rightarrow j$ adds i to j 's follower stock and makes i a potential future amplifier of j .

Reciprocal effect

If j follows back, $j \rightarrow i$ produces z_i^{+j} and the pair becomes mutual.

Eligible users

$P_{ij}^F(i \leftrightarrow j \mid i \nleftrightarrow j)$ is estimated among no-tie account pairs in which i follows j .

◀ Back to choice

Appendix: related literature — machinery, and our turn

The field models the *sharing* of misinformation; we model its *production* (life-cycle framing: Aridor, Jiménez-Durán, Levy & Song 2024)

- **Sharing games / echo chambers** — Acemoglu, Ozdaglar & Siderius (2024): veracity an *exogenous* draw; islands from platform network design.
Ours: veracity is *chosen*; islands emerge from differentiation under a content-blind gate.
- **Moderation** — Kominers & Shapiro (2024); Bar-Isaac, Deb & Mitchell (2025): robust removal; moderation as a priced product.
Ours: whack-a-mole is the spatial twin — removal *relocates* supply across niches.
- **Audience-building** — Gong (2021): niche launch recruits “zealot fans”; Filippas, Horton, Lipnowski & Parasurama (2025): attention bartering — follows exchanged for follow-backs; the follower stock is the asset.
Ours: price the fringe *location* in fabrication — niche entry becomes a theory of misinfo supply.
- **Engagement ranking** — Cunningham (2023): low quality wins the *vertical* score.
Ours: recruitment is *horizontal* — community-gated, not vertical convergence.

◀ back to the illustrations

Appendix: related literature — empirics, and where we sit

Empirical benchmarks — all on the *sharing* of exogenously given content:

- Allcott & Gentzkow (2017) — fake-news definition + 2016-election exposure.
- Vosoughi, Roy & Aral (2018) — falsehood spreads farther / faster (126k cascades); per-cascade reach, not the producer.
- Allcott et al. (2020); Martin, Vasserman & Pfiffer (2026) — deactivation RCT + structural news demand: the engagement–welfare wedge.
- Overgoor, Benson & Ugander (2019) — edge formation as discrete choice (our conversion frame).

Our setting: one high-salience cycle from the **supply side** — hand-labeled content, five event-level engagement modes, and the follow / unfollow graph.

Prior work's primitives — *who* produces falsehood, *who* amplifies, *who* profits durably — become our **outcomes**.

◀ back to the illustrations

Appendix: content classes — archetypes and examples

Classes describe the falsehood's *nature*, not the producer's intent

Class / archetype	What is false	Representative example (this cycle)
Distortion — false by design (deliberately misleading)		
Fringe theory	the story — an invented conspiracy or hidden cause	<i>"Bill Gates urges prep for smallpox attacks; a secret virus exercise"</i> @ichudov
Sensationalized claims	the magnitude — a true fact, overstated	<i>"Record daily cases — a clear and present danger, nearing a potential pandemic"</i> @DrEricDing
Fabricated sources / evidence	the backing — false authority or fake / relabeled evidence	<i>"Dr. Robert Malone on monkeypox"</i> (fake expert); a vaccine-reaction photo relabeled as mpox
Error	an honest slip on fact or timing; a correction resolves it	<i>"Ghana records its first monkeypox case"</i> — but it hadn't; refuted by the Ghana Health Service @AfricaStoryLive
Satire	nothing hidden — false on its face, as a joke	<i>"Health experts soon: wearing a monkey mask protects from monkeypox"</i> @DrEliDavid

Each is the human **ground truth** — adjudicated by reading the replies, the source data, and the official record, *not* by matching a fact-check article (which would miss most of them).

Appendix: recurring templates, and what makes labeling hard

Five templates carry most of the misinfo

1. **Planned / bioengineered** — by Gates / WHO / government (for profit, fear, control, depopulation).
2. **Airborne** transmission, downplayed by officials.
3. **Same as** shingles / chickenpox / smallpox.
4. **Vaccines cause harm** (a covid-vaccine side-effect; smallpox jab causes AIDS).
5. **Authorities are deceiving you** — severity dialed up or down, information suppressed.

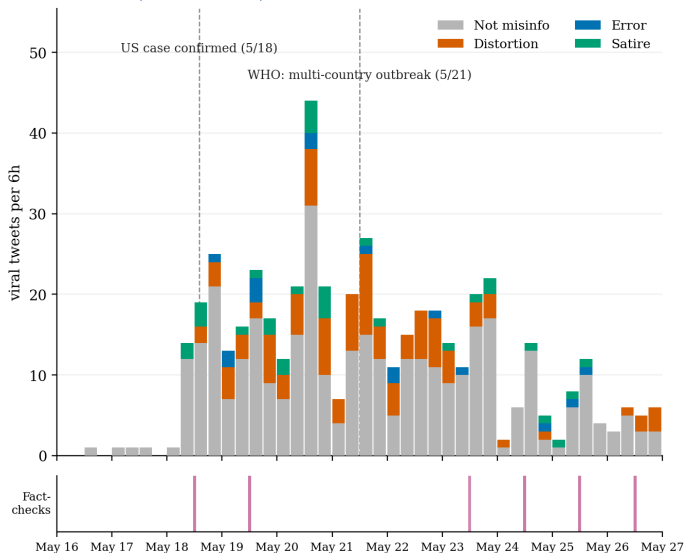
Templates recur from the COVID cycle — the location is recycled, only the disease changes.

Why only a human gets these right

- **Laundering**: a discredited expert lends authority — *“Dr. Robert Malone / James Delingpole on monkeypox.”*
- **Complexity-clad**: real jargon overwhelms the reader (@Kevin_McKernan, @Jikkyleaks).
- **True-but-deceptive**: *every fact checks out; the frame is the lie* — the 2021 NTI/Munich wargame, posted as “coincidence?”
- **Satire inverts**: a joke tweet may *quote* misinfo to mock it — its message is the opposite of its content.
- **The machine’s mirror error**: describing the conspiracy (“#BillGatesBioTerrorist trends”) gets flagged as the conspiracy.

Appendix: the onset by content class

Calendar time, 6-hour bins; 503 viral tweets in the window

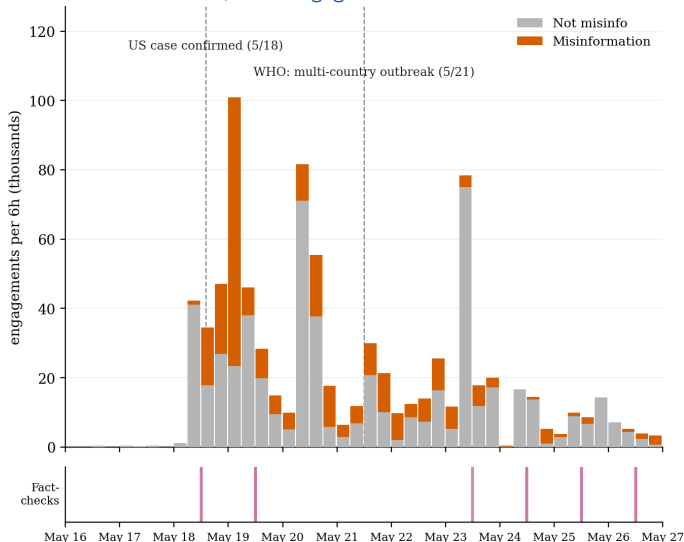


- All three classes appear in the first burst: on May 18 the day's misinfo is already **distortion**, **error**, and **satire** together.
- **Distortion** dominates throughout — 103 of the 150 misinfo tweets in the window — and carries the peak.
- **Error** (16) and **satire** (31) stay thin at every point of the cycle.

▶ back

Appendix: the onset weighted by engagement

Same window and bins; 834k engagements instead of 503 tweets

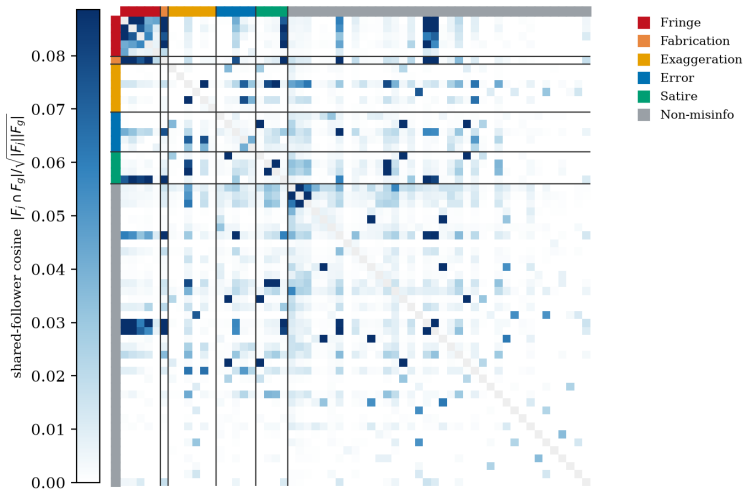


- Over the window, engagement tracks volume: misinformation takes 33% of engagement against 30% of tweets — $1.14\times$ per tweet.
- **The burst is where it separates:** misinformation carries 77% of the engagement in the peak 6-hour bin (May 19).
- It then **fades faster than it posts:** from May 23 its engagement share runs below its share of tweets.

Engagement = distinct retweeters, repliers and quoters recorded at tweet level; likes are not recorded for the corpus, and the counts undercount — read as a floor.

Appendix: audience overlap — exact 59×59 cosine

Audience overlap: exact shared-follower cosine, all 59 authors

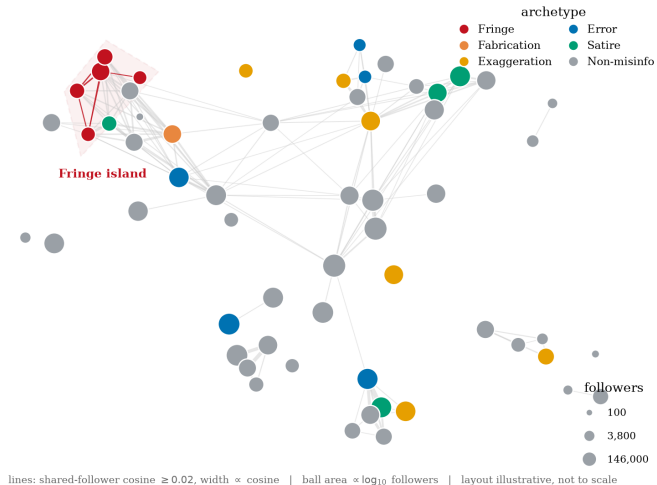


Each cell is the exact shared-follower cosine $|F_j \cap F_g| / \sqrt{|F_j| |F_g|}$; authors ordered by archetype block. The fringe block (top-left) is the densest; mainstream-news pairs supply the largest single overlaps.

[◀ Back to overlap](#)

Appendix: audience-overlap network

Fringe authors' audiences overlap most with each other

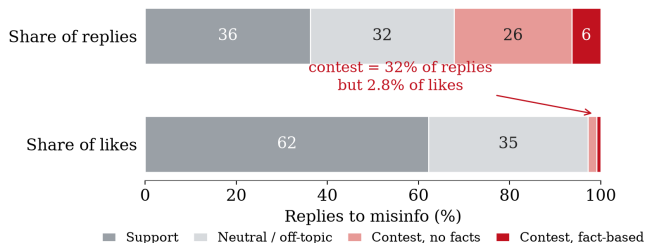


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Appendix: stance shares, reply-weighted vs like-weighted

The crowd does not amplify contestation

n=7,795 misinfo replies with like counts; likes are current-day totals

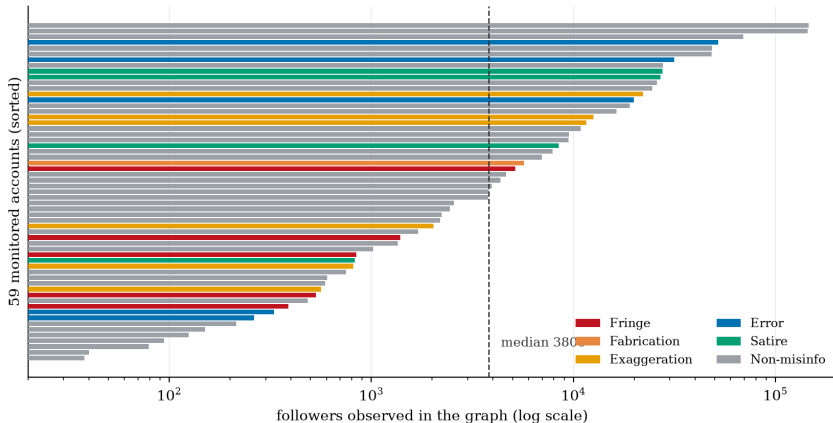


- Weighting each misinfo reply by its likes: contest falls from 32% of replies to 2.8% of likes; support rises to 62%.
- Like counts are current-day totals and tail-driven (median reply ≈ 0 likes).

◀ back

The follow network we observe

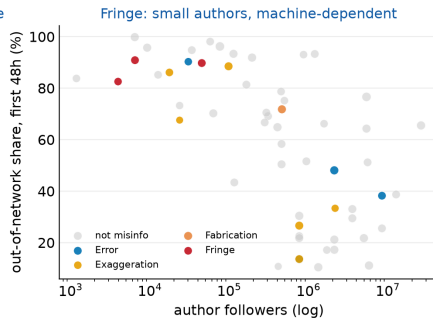
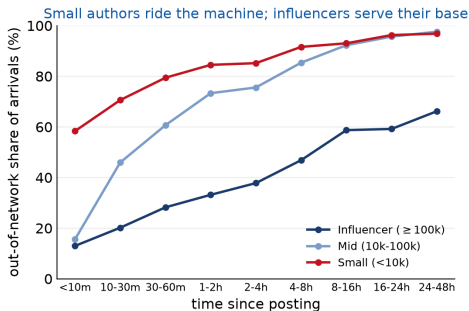
We observe 746,325 followers of 59 news actors, with follow/unfollow times



- 59 monitored accounts, 746,325 distinct followers; audiences span three orders of magnitude (median 3,806, max 146,193) — **size does not track archetype**.
- Every follower carries a **follow / unfollow timestamp**, so follow-state is known at the moment of each engagement.

Who needs the machine: influence, and archetype

Pre-existing audience determines whether reach is served in-network or rented from the gate



- **Influencers ($\geq 100k$) serve their base at once** (first hour 79% in-network; 48% of 48h reach out-of-network); **small authors ($<10k$) get 94% of reach through the gate** (first hour 25%).
- Machine-dependence is monotone across archetypes: **fringe 88%** (median author 6,449 followers) > fabrication 72% > sensationalized 51% > **error 44%** (news, 2.1M median) — no audience of your own $\Rightarrow$ **rent the machine**. (Fabrication: one panel tweet.)

The asset, and the right outcome variable

- In 2022 there is **no ad-revenue share** (it arrives July 2023); engagement per se earns nothing.
- A producer's **only monetizable asset is the follower stock** — converted off-platform, $M = m F$.
- So the payoff to a tweet is the **conversion of a non-follower into a follower**, not the engagement it collects.

The right outcome is **follower yield**, not engagement — and the unit is the audience, not the post.

Fact 1: attention converts to audience — misinfo more so

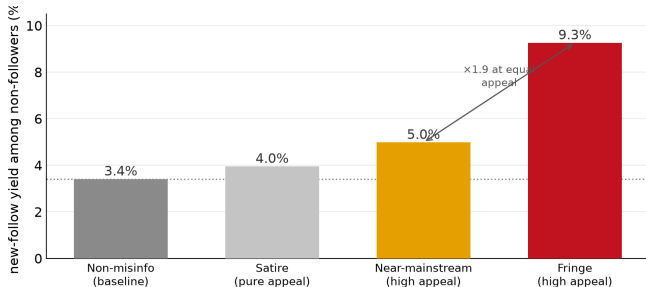
59% of engagements are by **non-followers** of the author — engagement is reach, not preaching to the choir.

- Among non-followers, engaging **misinfo** converts to a follow at **7.4%** vs. 3.4% for accurate content (2.2×).
- Posting misinformation is followed by a follower spike.

Both a vertical (engagement-ranking) and a positioning story predict this. To separate them, look *where* the premium sits.

Fact 2: the premium is fringe-*location*-specific, not appeal-specific

Refutes a single vertical engagement axis (Cunningham et al. 2023)

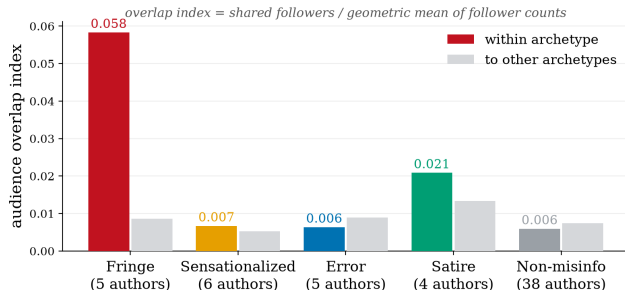


- At **equal high appeal**, fringe location converts 9.3% vs near-mainstream 5.0% ($1.9\times$).
- **Pure appeal converts nobody:** satire — maximally viral, no fringe location — sits at the 3.4% baseline.

A single vertical axis cannot make a location gap at fixed appeal.

Audiences cluster by content location, not by size

Audience overlap concentrates in the fringe



- **Fringe authors share audiences with each other** — within-group overlap 0.058 vs 0.009 to any other archetype (6.8×), the sharpest clustering in the network.
- Overlap is otherwise sparse (median author-pair shares 2 followers); the few dense overlaps are mainstream-news pairs (up to 32,768).

A fringe follow buys a distinct community, not a bigger audience — the horizontal location is visible in the follow graph.

► Exact heatmap

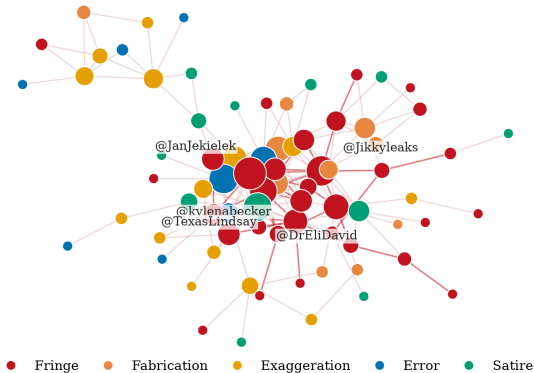
► Network map

... and the producers follow *each other*

Reciprocal follows scraped from author following-lists — the community is deliberate

Misinformation authors are a mutual-follow community --- fringe at the core

193 mutual-follow pairs (45 fringe ↔ fringe); protected accounts censored

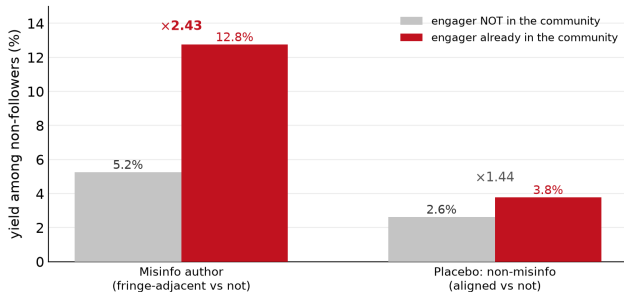


- **45 fringe ↔ fringe mutual-follow pairs** — fringe authors explicitly follow one another, with fabrication and sensationalized authors attached at the edge.
- A **censored floor**: 27 fringe accounts are protected, so their outgoing follows are unobserved.

Not just shared audiences (the cosine map) — an explicit, author-to-author community.

Fact 3: conversion is community-gated

Refutes demand-gullibility and a community-blind vertical score

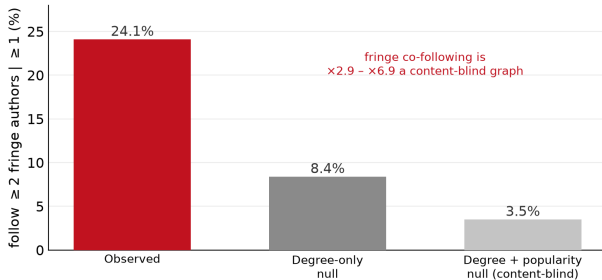


- Engagers already **fringe-adjacent** convert at 12.75% vs 5.25% ($2.43\times$).
- Far above the **generic-homophily placebo** — following a misinfo author then converting to a *non-misinfo* author is only $1.44\times$.

Recruitment lands where the viewer already sits in the community — a horizontal position, not a gullibility level.

Fact 4: the community assembles follow-by-follow

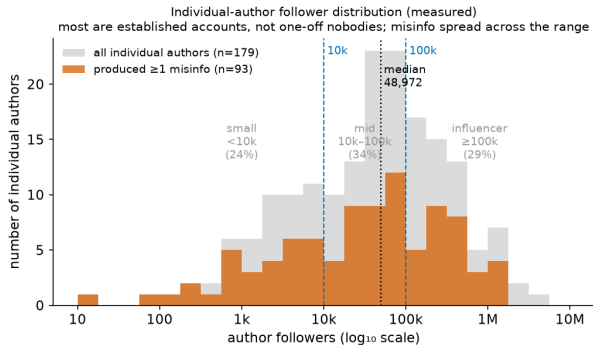
Refutes platform-*designed* echo chambers (Acemoglu, Ozdaglar & Siderius 2024)



- Engagers who follow one fringe author disproportionately follow **two** — $2.9\times$ – $6.9\times$ a content-blind graph.
- Accretion is **slow and spaced** (sequential $1.5\times$), not algorithmic same-day bundles.

Islands emerge from producer differentiation under a content-blind gate — not from reliability-contingent platform design.

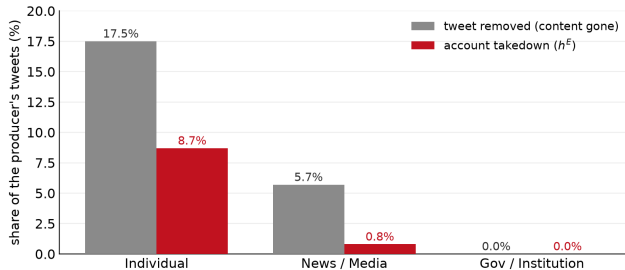
Who supplies it — full size range, and mostly fringe



- Misinfo authors span **small to influencer**; skew to **individuals** ($P(\text{misinfo}) = 0.46$ vs. 0.13 news; $6.0\times$ odds).
- **Fringe conspiracy is the modal type** of misinformation (65 tweets — the largest archetype).

Within a producer, a misinfo tweet also earns $1.59\times$ the retweets (account FE) — but that is equally a *vertical* fact; the discriminators come next.

Two moderation margins hit two different populations

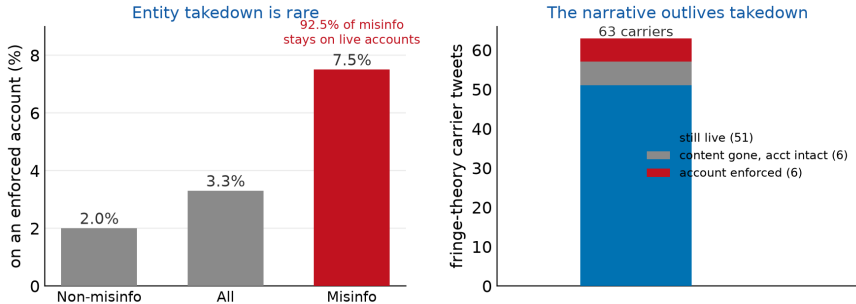


- **Account takedown** (h^E , taxes the follower *stock*): 86% individual accounts, concentrated in distortion / fringe.
- **Content removal** (h^C , taxes the recruit *flow*): broad, dominated by benign news self-retraction.

The model's $h^C \neq h^E$ split is in the data.

(Only account takedown is cleanly identified, $n=28$; self- vs platform tweet removal is not separable.)

Punishment is weak; the narrative outlives the takedown



- Entity punishment is **rare** — 92.5% of misinfo stays on live accounts (corroborates weak punishment / small h^E).
- Structural action is a **sharper** signal (OR 4.0) than content removal (OR 1.9, contaminated by news self-correction) — *consistent with* Kominers & Shapiro.
- Removal touches a fraction of a narrative's carriers (6 of 63) — *consistent with* whack-a-mole (re-emergence is outside this cross-section).